Lecture 10

Red-Black Trees: Deletion (contd.), Augmenting Data Structures

Source: Introduction to Algorithms, CLRS

Cases for Deletion

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There are four cases when deletion causes only **Property 6's** violation:

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- **Case 1:** x's sibling w is red. → *Case 1 will convert to Case 2, 3, or 4*
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Case 2 will either terminate or move x towards the root

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Case 3 will convert to Case 4

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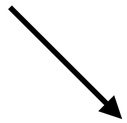

Case 4 will terminate

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What if x or its sibling w is NIL?

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Idea: Give x an **extra black colour** that it will lose without violating RB-tree properties.

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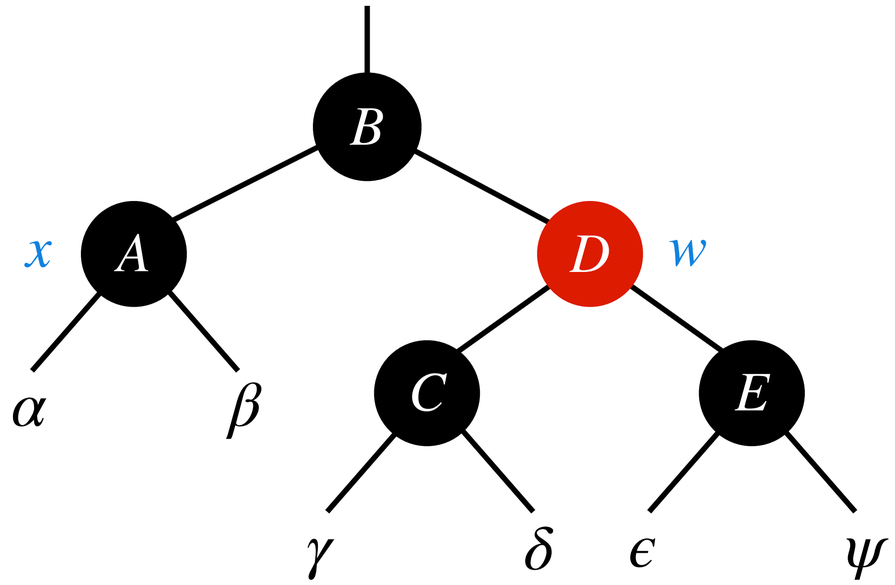

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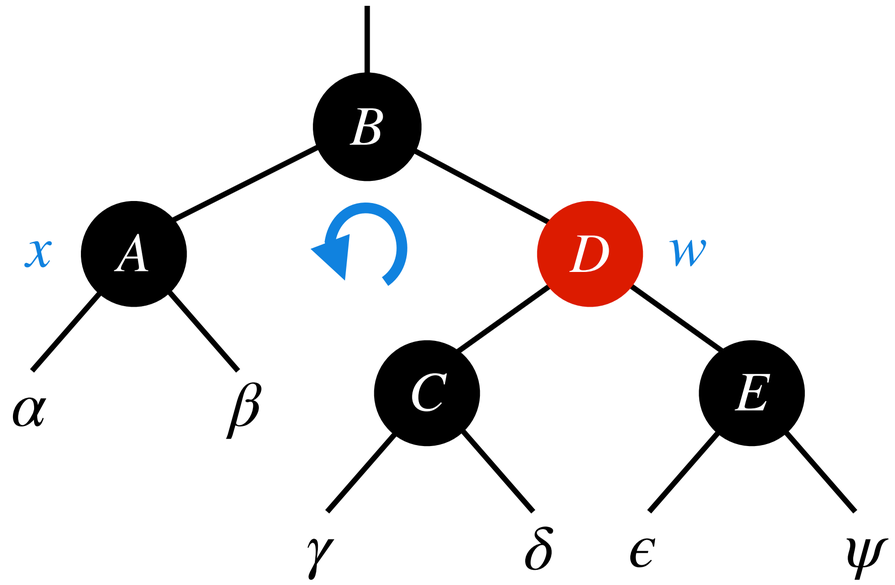

Handling: Switch colours of w and $x\,.\,p$ and perform a left rotation at $x\,.\,p$.

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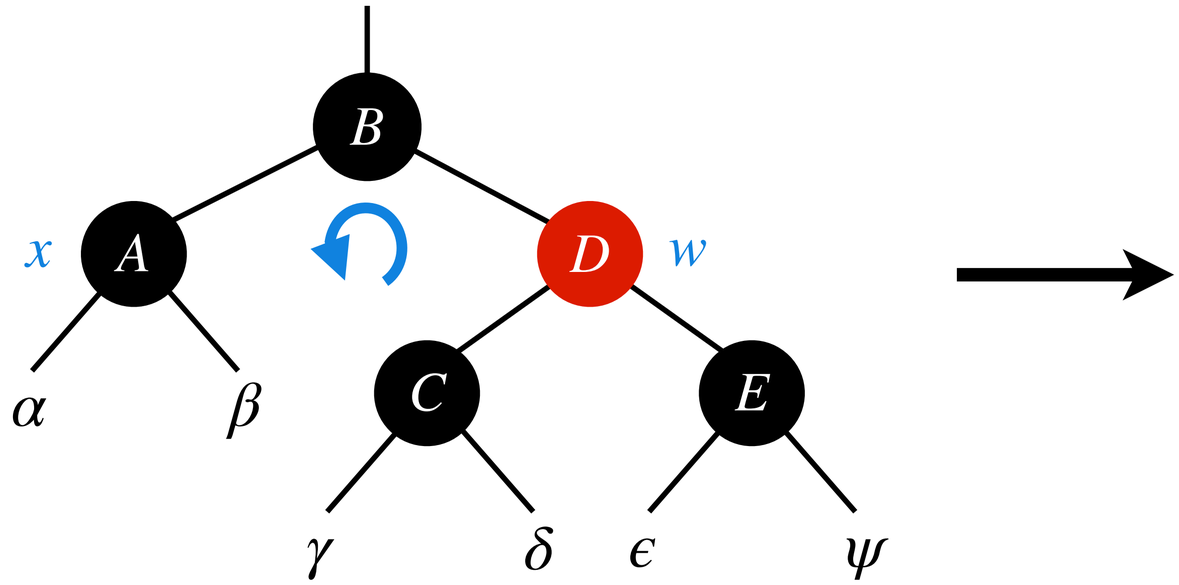

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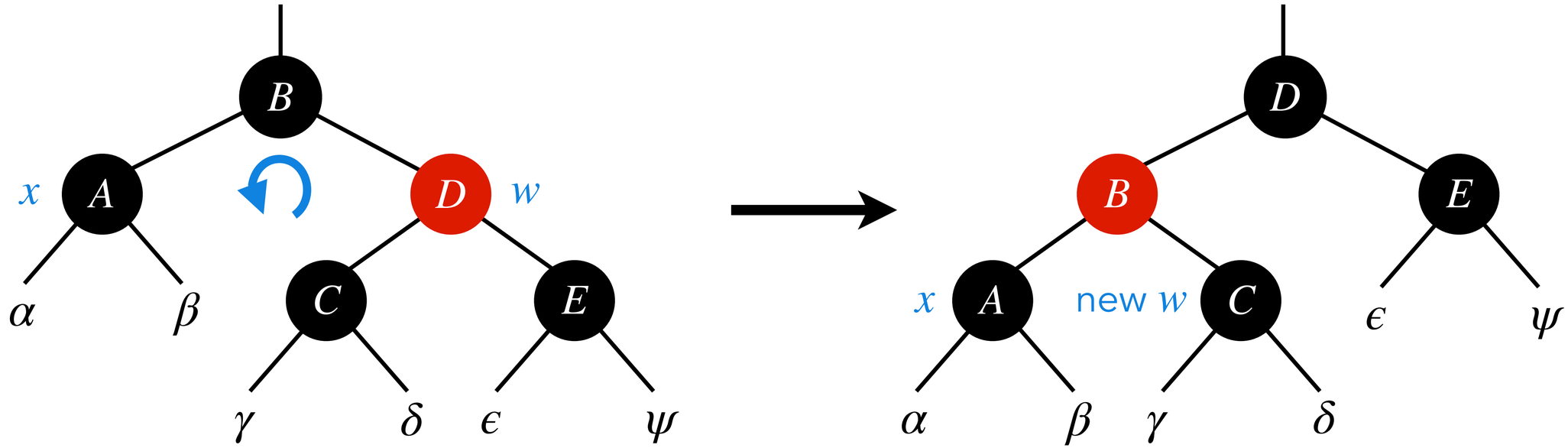

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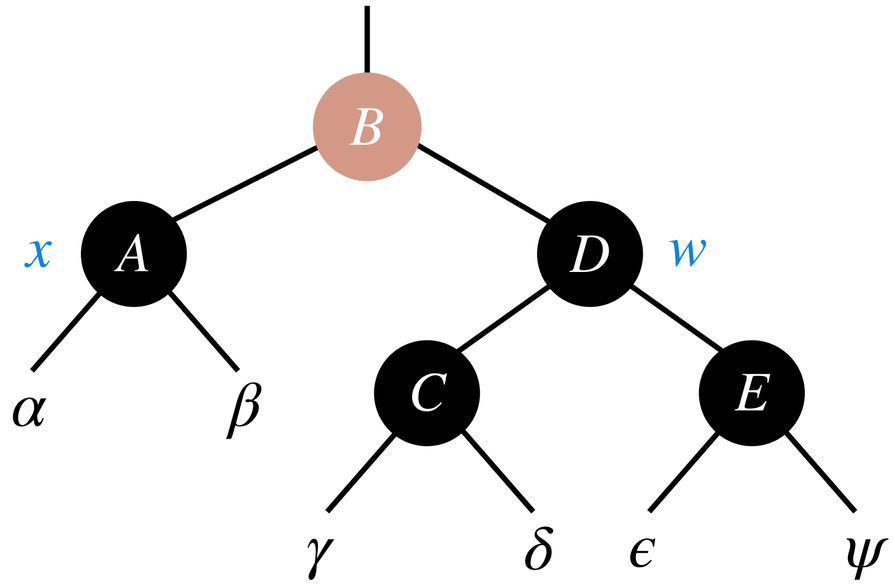

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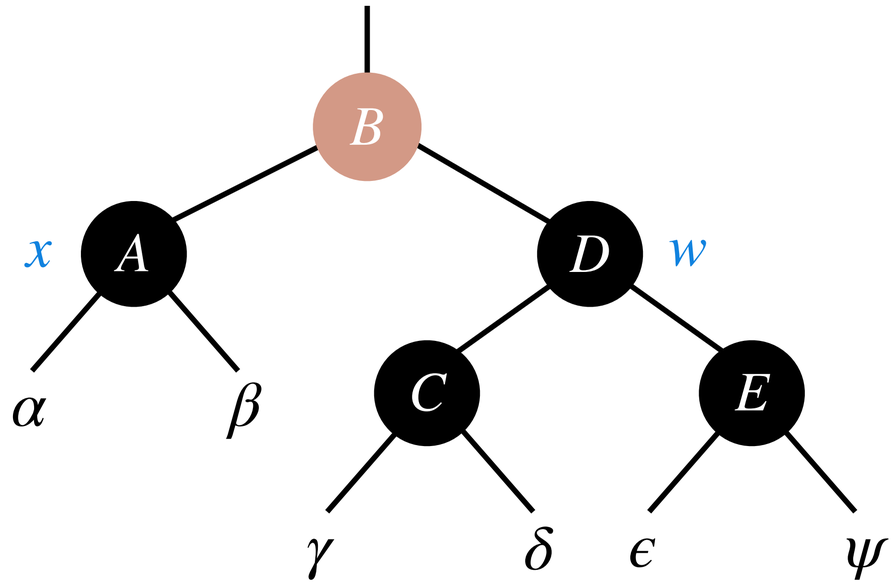

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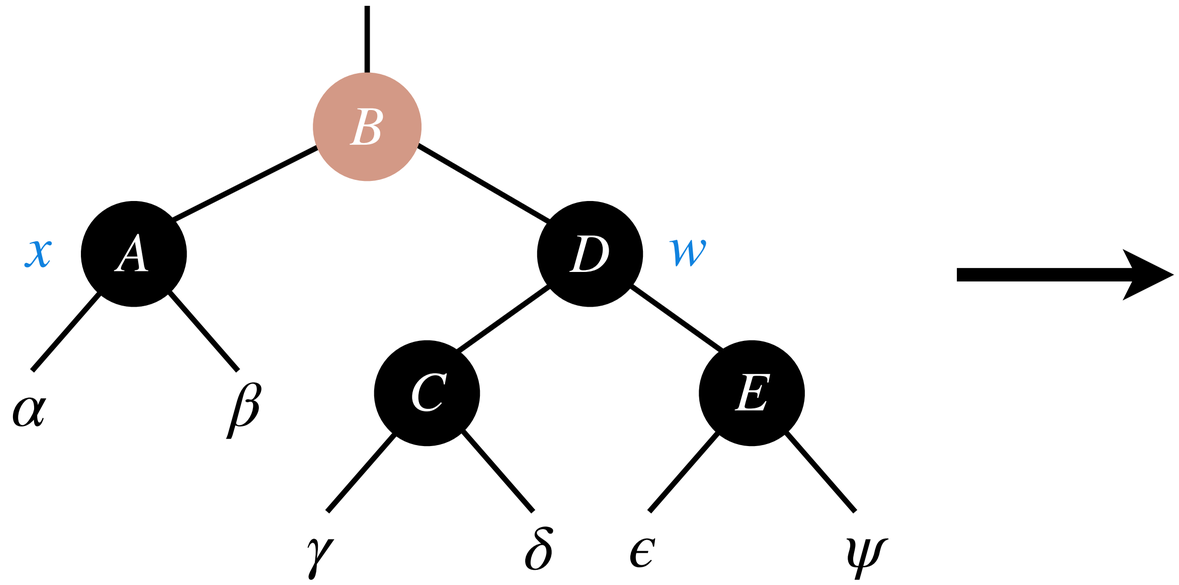

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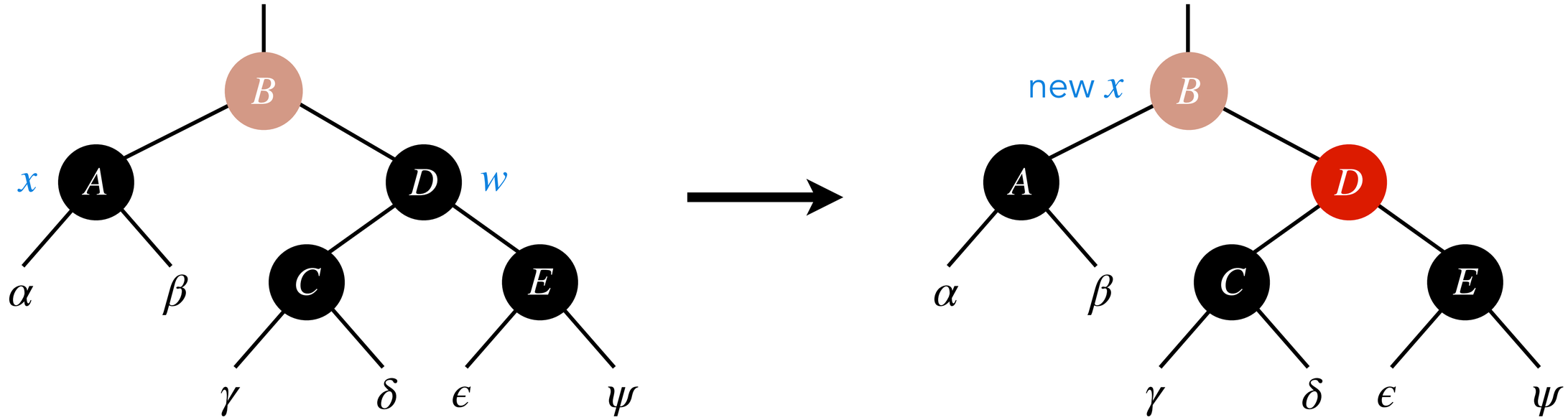

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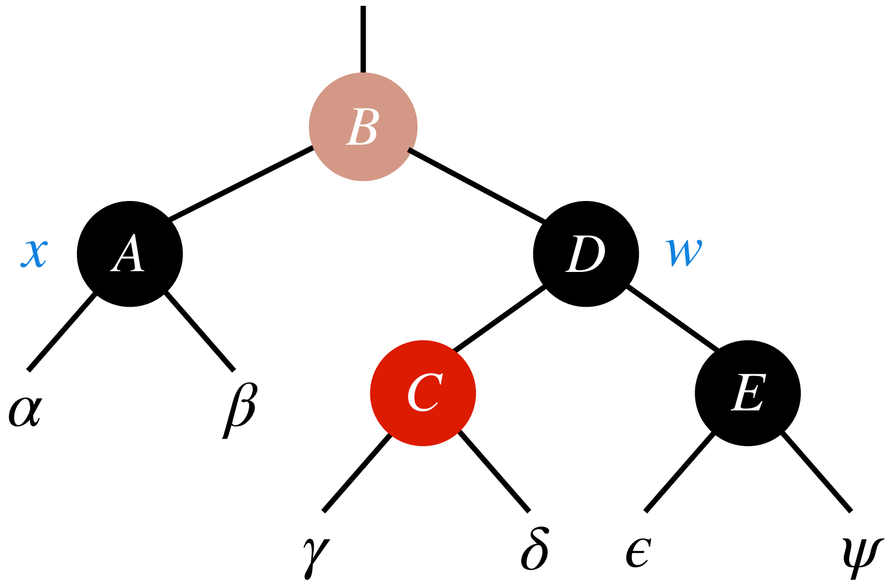

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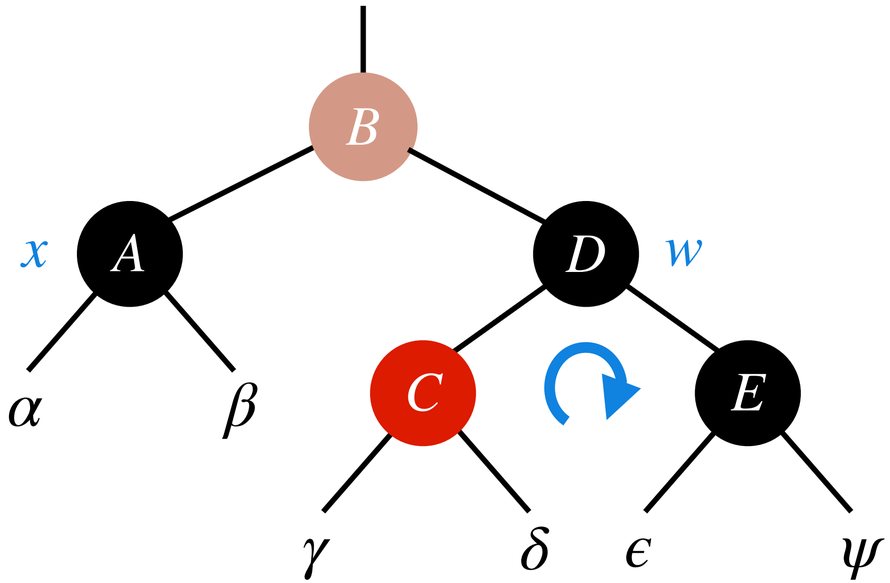

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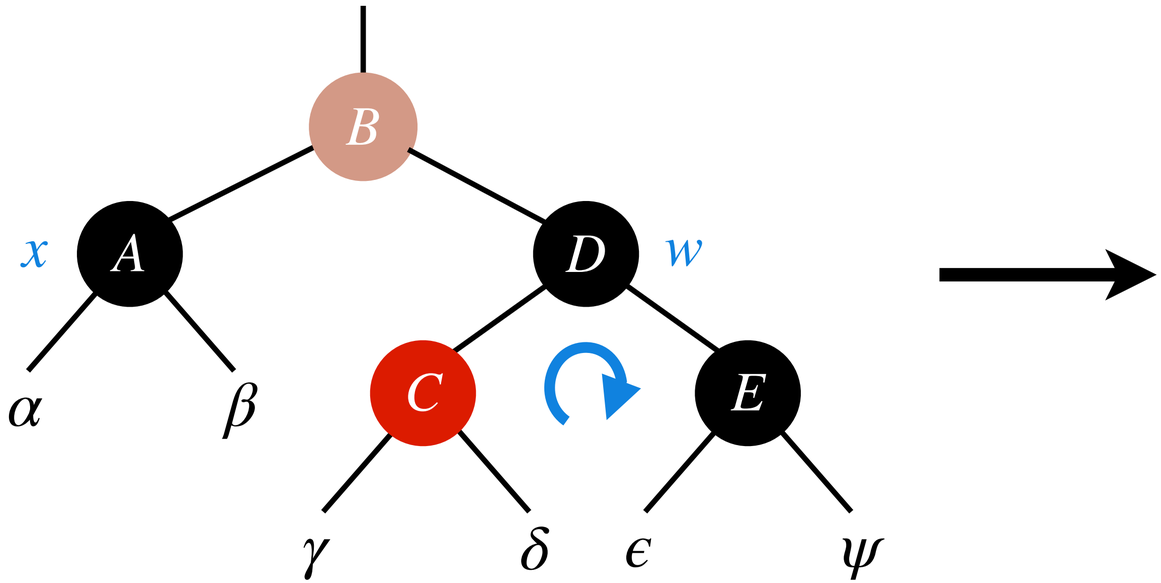

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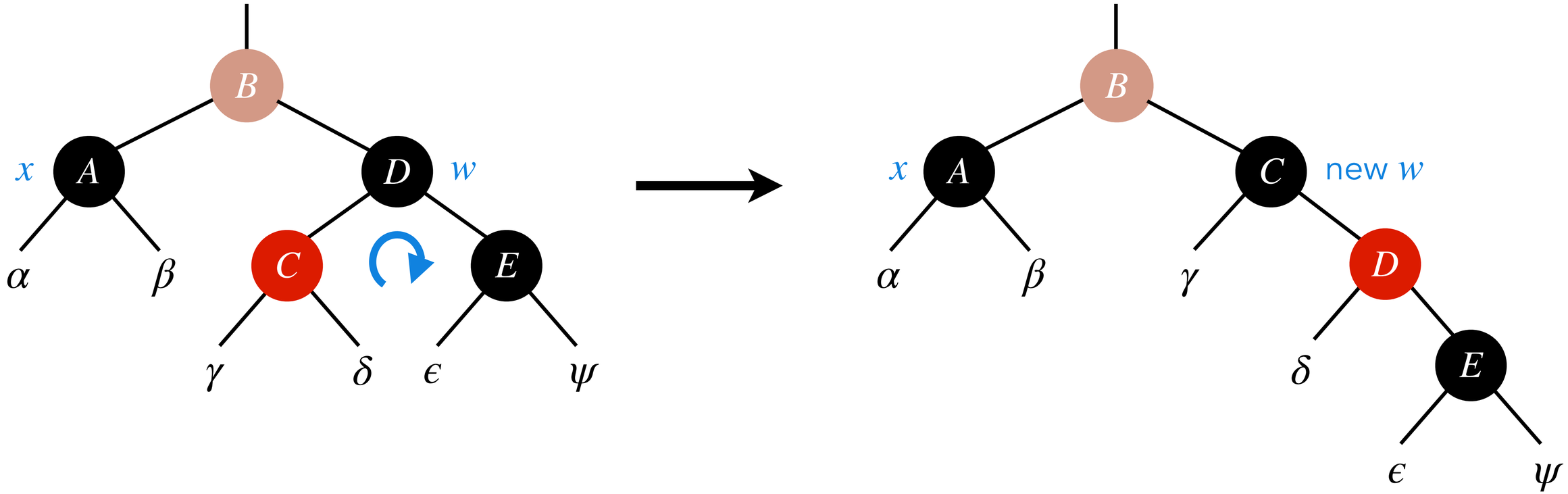

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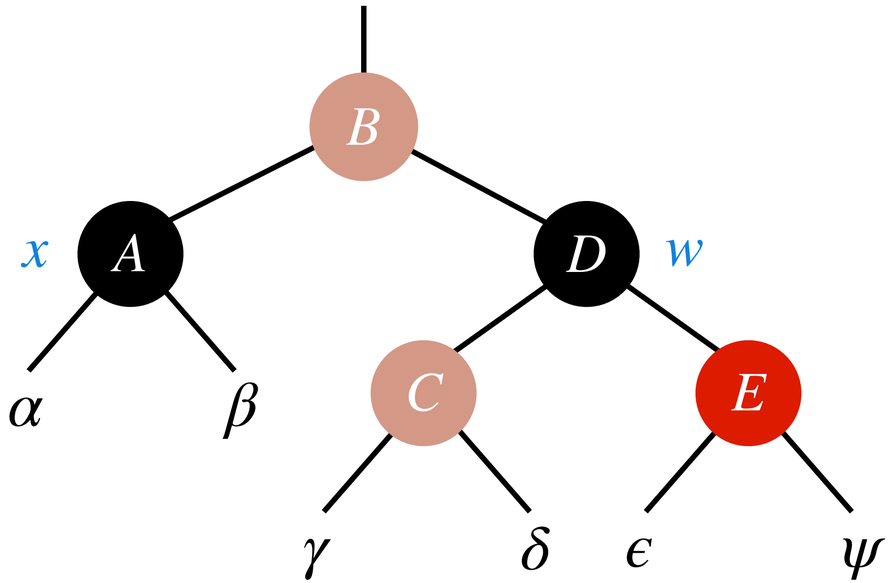

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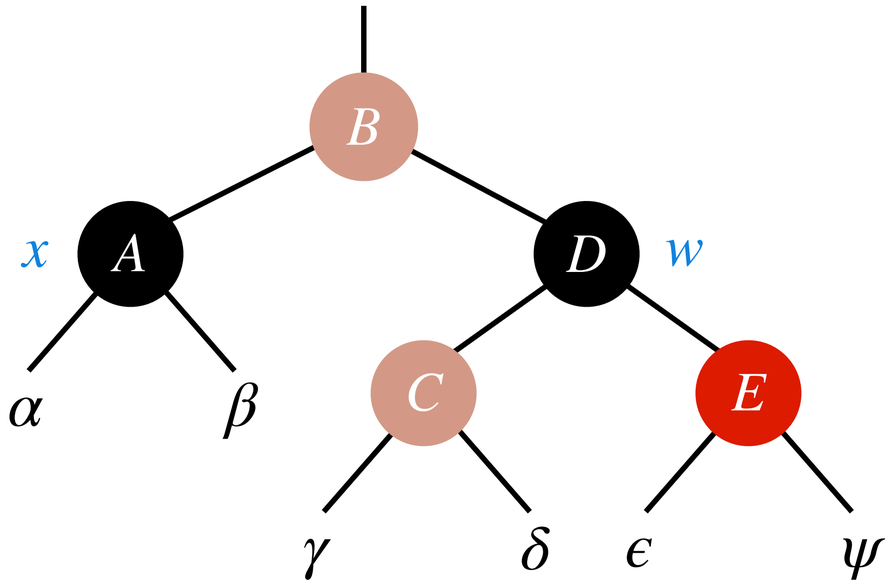

Handling: Give w the colour of x's parent. Make x's parent and w's right child black.

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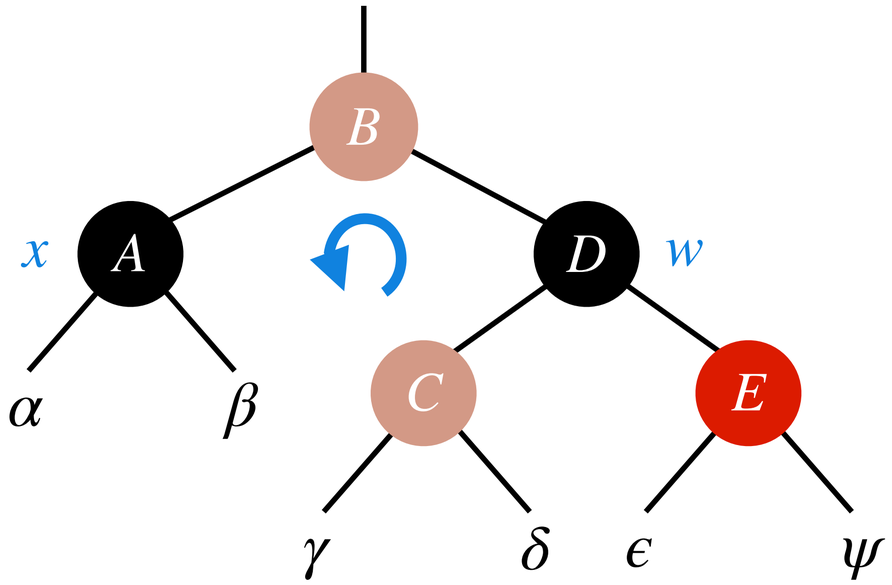

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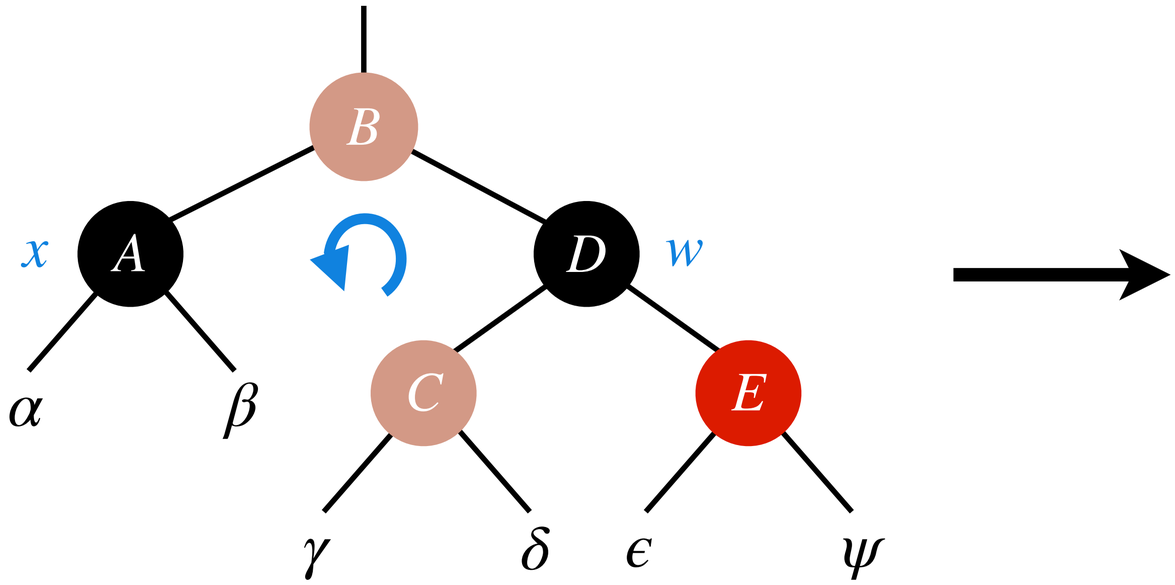

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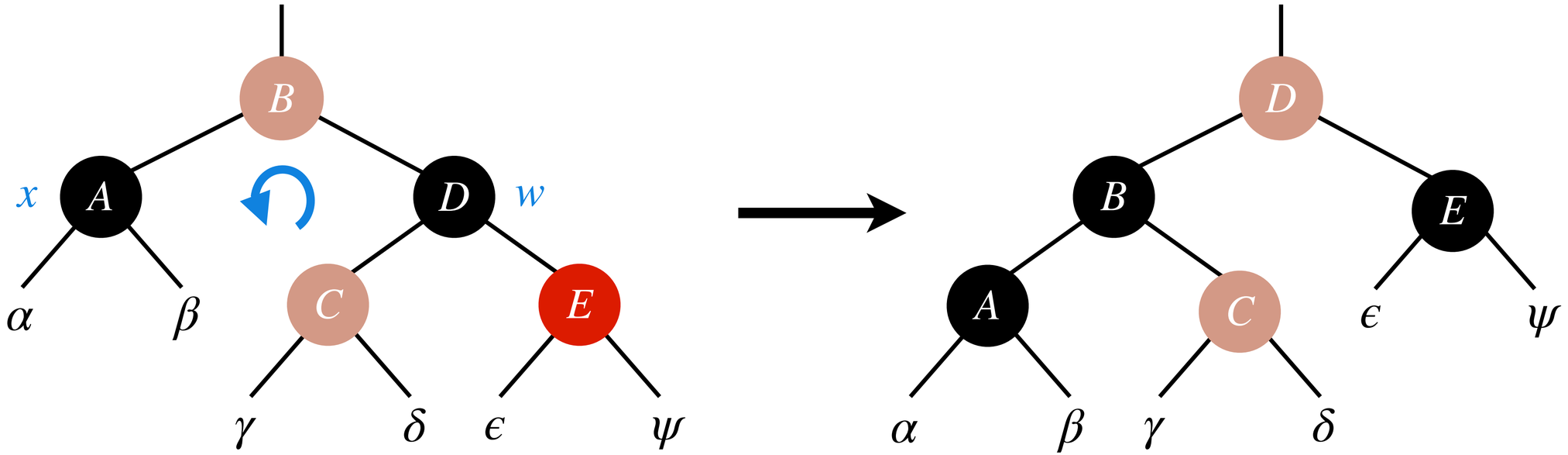

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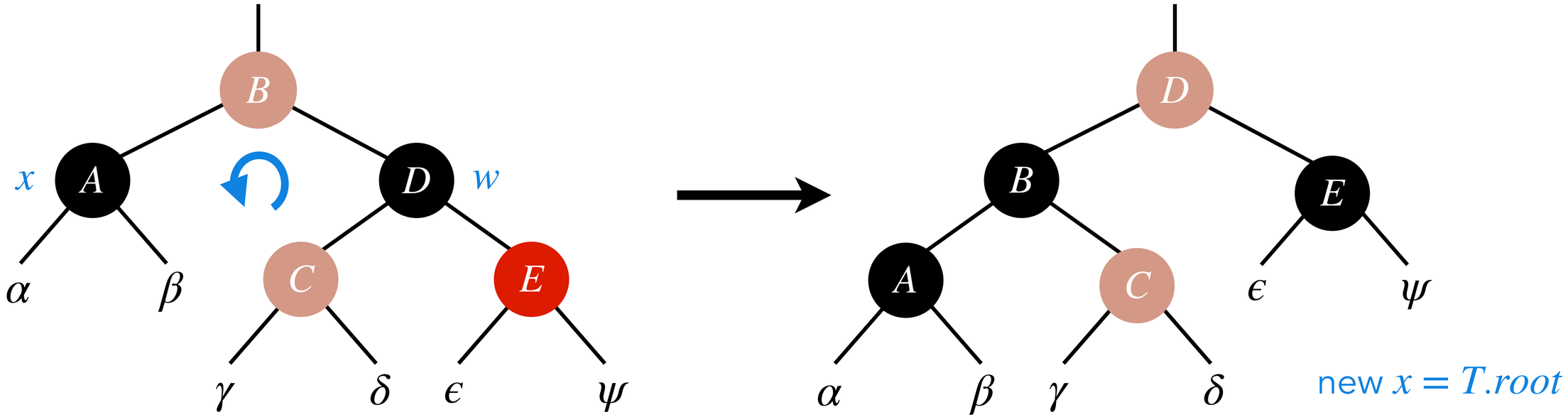

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Defn: In a set, rank of an element is its position in the sorted order (w.r.t. keys) of the set.

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Elements with ranks 1, 3, and 8 are 5, 10, and 40, respectively.

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Ranks of elements 5, 15, and 25, are 1, 4, and 6, respectively.

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Suppose we want to maintain a dynamic set with the following operations:

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Contains elements with keys.

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Suppose we want to maintain a dynamic set with the following operations:

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Should we learn or invent a new data structure to facilitate these operations?

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No. An easy modification to RB-trees is enough.

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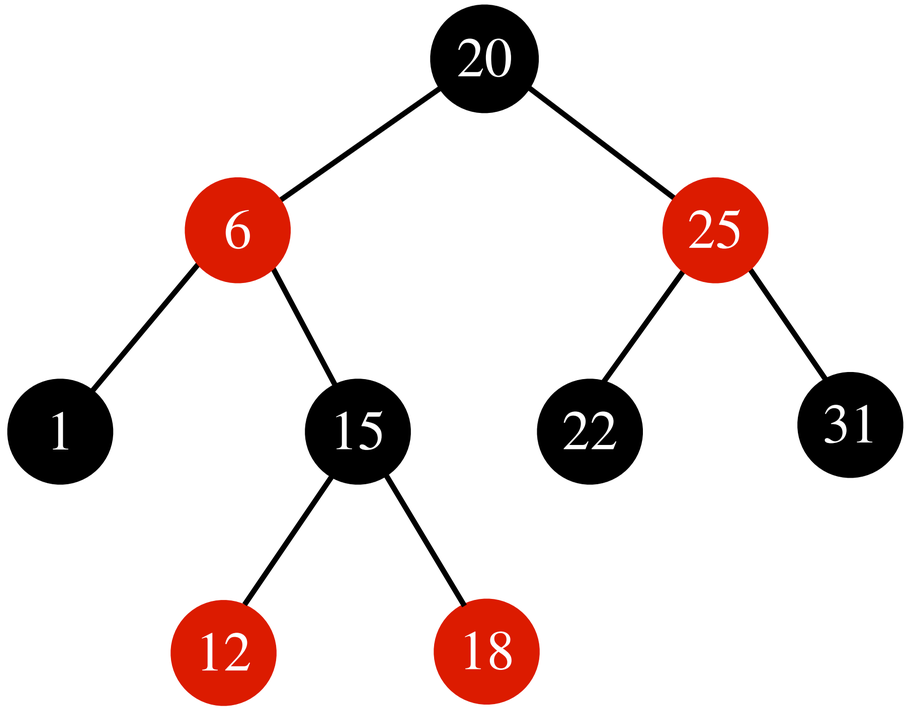

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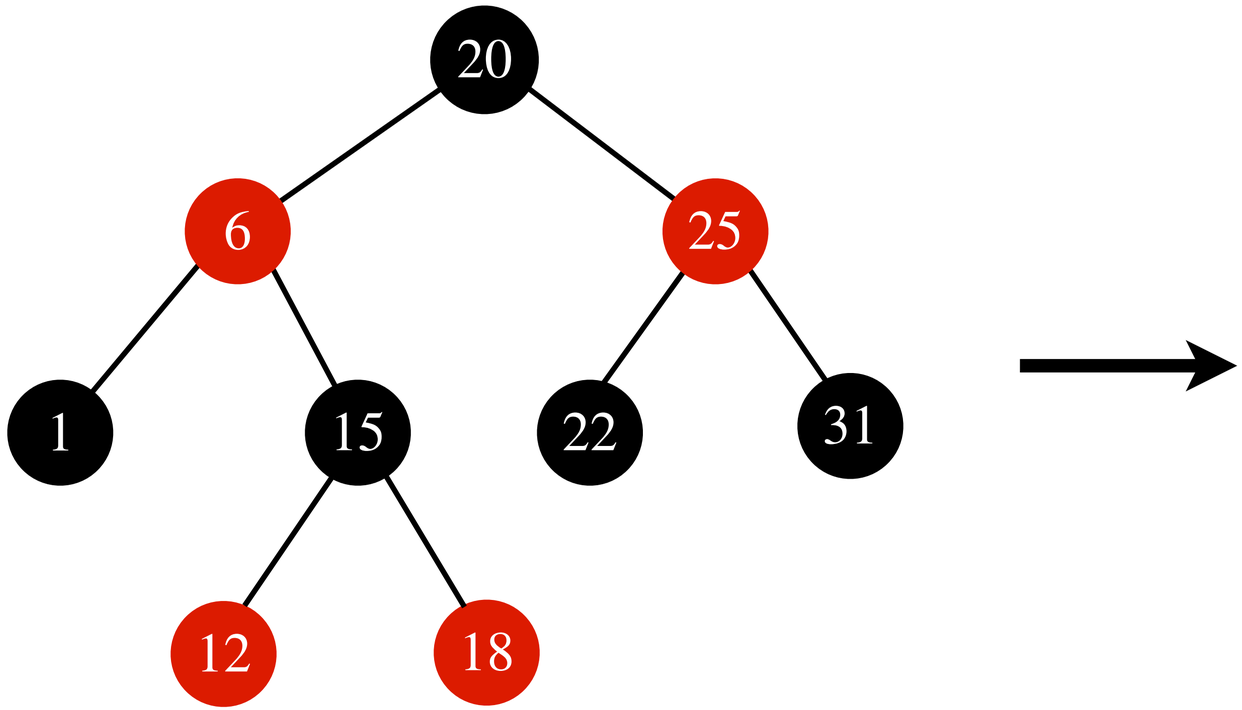

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